EXTERNAL APPLICATIONS OF PRIME-CIRCLE METHODS

Toeplitz Counterexamples and Mertens Mean Oscillations

An exact conjecture refutation and a convex-envelope argument

Research prepared for Keio

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Abstract. We apply two structures developed in a prime-circle investigation to questions posed independently in the literature. First, the canonical strongly convex function of order $0 < \alpha < 1$ used in Kowalczyk–Lecko–Śmiarowska (2021) has fourth Hermitian Toeplitz determinant

$$(1-\alpha^2)^3\left(1-\frac{\alpha^2}{324}\right).$$

It therefore refutes the lower bound in their Conjecture 1. At $\alpha = 1/2$, an exact rational witness gives $1295/3072$ against the claimed lower bound $1296/3072$. Toeplitz prediction-error monotonicity propagates the violation to every matrix order $q \geq 4$. Second, a slope-selection lemma for a concave lower envelope of zero-dependent affine functions, combined with Zhao's published oscillation and second-moment lemmas, removes the smooth-boundary assumption in the hypothetical $\Theta = 1$ case of his Mertens-product mean theorem. The prime-power tail is retained exactly, and the other cases are joined with explicit estimates to complete the original mean-value characterization of RH. The constructions, quantifiers, and imported inputs are stated explicitly. The work does not prove RH. Classical moment positivity and convex-envelope techniques are credited as such; historical priority of the particular applications is not certified by the bounded literature search.

Exact finite certificate

$$\det\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} & \frac{7}{48} \\ \frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{7}{48} & \frac{1}{4} & \frac{1}{2} & 1 \end{pmatrix} = \frac{1295}{3072} < \frac{1296}{3072} = \left(1-\frac{1}{4}\right)^3.$$

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1 Two original targets and the transfer mechanisms

The companion manuscript *Prime Reciprocal Circles and the Riemann Zeta Function: A Unified Energy, Operator, and Sign Theory* developed finite Toeplitz energies and one-sided arithmetic moments. The present paper uses those viewpoints on two externally posed questions. All arguments needed for the transfers are reproduced here, apart from clearly cited classical theorems and the two analytic estimates stated in Section 3.1.

Original target	Mechanisms combined	Result and scope
Kowalczyk et al., Conjecture 1 (2021)	A disk-to-sector map, exact Taylor coefficients, reflection blocks, and Toeplitz prediction errors	The original lower bound is false for every $0 < \alpha < 1$ and every $q \geq 4$. A rational 4×4 witness already proves the refutation.
Zhao, Section 8, Question (1) (2025)	A lower envelope indexed by actual zeros, concavity, local oscillation, and an upper second-moment bound	The difficult $\Theta = 1$ branch has unbounded mean oscillations in both signs without the proposed smooth-envelope assumption.

The first statement changes neither the analytic class nor the constant in the proposed inequality. The second concerns the original Mertens product and its ordinary integral, not a modified sensor. In both cases the mathematical conclusion is separate from the question of historical priority. In particular, the general theory of positive Toeplitz matrices and lower envelopes of zero lines predates this work.

What is not asserted. No finite computation is used as evidence for RH. We do not claim a sharp replacement lower bound for the strongly convex class. The remaining statements about RH are equivalences or implications proved using the specified established inputs. The source/status audit at the end records the scope of the proofs, the imported estimates, and the limits of the historical search.

2 An exact counterexample to a strongly convex Toeplitz conjecture

The circle-moment viewpoint has an application to an independently published conjecture. Kowalczyk, Lecko and Śmiarowska [1, Conjecture 1, p. 331] proposed that

$$(1-\alpha^2)^{q-1} \leq \det T_{q,1}(f) \leq 1, \qquad 0 < \alpha \leq 1, \quad q \geq 2, \tag{2.1}$$

for every normalized strongly convex function of order α. The statement is false for every $0 < \alpha < 1$ and every $q \geq 4$. The first failure has a rational certificate of size four. The counterexample below is the very function used in [1, Eq. (21), p. 329] to attain their proved low-order bounds. The conclusion does not identify an optimal replacement lower bound.

2.1 Definitions and an admissible analytic witness

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, and normalize $f(z) = z + \sum_{n\geq 2} a_n z^n$. Its Hermitian Toeplitz matrix is

$$T_{q,1}(f) = \begin{pmatrix} 1 & a_2 & a_3 & \cdots & a_q \\ \overline{a_2} & 1 & a_2 & \cdots & a_{q-1} \\ \overline{a_3} & \overline{a_2} & 1 & \cdots & a_{q-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \overline{a_q} & \overline{a_{q-1}} & \overline{a_{q-2}} & \cdots & 1 \end{pmatrix}.$$

The class $\mathcal{C}_\alpha$ in [1] is defined by

$$\left|\arg\left(1+\frac{zf''(z)}{f'(z)}\right)\right| \le \frac{\alpha\pi}{2} \qquad (z\in\mathbb{D}).$$

For $\alpha = 1$ this is the normalized convex class.

Fix $0 < \alpha < 1$, take the analytic principal logarithm on the right half-plane, and define

$$Q_\alpha(z) = \exp\left(\alpha \operatorname{Log}\frac{1+z}{1-z}\right), \tag{2.2}$$

$$H_\alpha(z) = \int_0^z \frac{Q_\alpha(t)-1}{t}\,dt, \qquad f_\alpha(z) = \int_0^z \exp(H_\alpha(w))\,dw. \tag{2.3}$$

The quotient in the first integral has a removable singularity at zero. Both integrals are primitives of holomorphic functions on the simply connected disk and are consequently path independent.

Lemma 2.1 (Membership in the original class)**.** *For every $0 < \alpha < 1$, the function f_α belongs to $\mathcal{C}_\alpha$ and has real Taylor coefficients. In particular, it is analytic, convex and univalent throughout $\mathbb{D}$, with $f_\alpha(0) = 0$ and $f'_\alpha(0) = 1$.*

Proof. The Cayley map $(1+z)/(1-z)$ takes $\mathbb{D}$ to the open right half-plane. Thus Q_α is analytic, $Q_\alpha(0) = 1$, and $|\arg Q_\alpha(z)| < \alpha\pi/2$. Differentiating (2.3) gives $f'_\alpha = \exp H_\alpha \neq 0$ and

$$1+\frac{zf''_\alpha(z)}{f'_\alpha(z)} = Q_\alpha(z).$$

The defining sector condition is satisfied, with strict inequality. The normalization follows directly from $H_\alpha(0) = 0$. Since $\Re Q_\alpha > 0$, the standard analytic convexity criterion also gives convexity and univalence. Conjugation symmetry of (2.2) and its primitives proves that the coefficients are real. □

2.2 A fourth-order defect detected by a Schur complement

Theorem 2.2 (Exact failure of the published lower bound)**.** *For every $0 < \alpha < 1$,*

$$\det T_{4,1}(f_\alpha) = (1-\alpha^2)^3\left(1-\frac{\alpha^2}{324}\right) < (1-\alpha^2)^3. \tag{2.4}$$

Consequently [1, Conjecture 1] is false.

Proof. Expanding the analytic logarithm and exponentials yields

$$Q_\alpha(z) = 1 + 2\alpha z + 2\alpha^2 z^2 + \frac{2\alpha+4\alpha^3}{3}z^3 + O(z^4),$$

$$H_\alpha(z) = 2\alpha z + \alpha^2 z^2 + \frac{2\alpha+4\alpha^3}{9}z^3 + O(z^4),$$

$$f_\alpha(z) = z + \alpha z^2 + \alpha^2 z^3 + \frac{\alpha+17\alpha^3}{18}z^4 + O(z^5).$$

Set

$$a = \alpha, \qquad v = 1-a^2, \qquad c = \frac{a+17a^3}{18}, \qquad d = c-a^3 = \frac{a(1-a^2)}{18}.$$

The matrix in question is

$$T_{4,1}(f_\alpha) = \begin{pmatrix} 1 & a & a^2 & c \\ a & 1 & a & a^2 \\ a^2 & a & 1 & a \\ c & a^2 & a & 1 \end{pmatrix}.$$

Its top-left block $A = T_{3,1}(f_\alpha)$ satisfies

$$\det A = v^2, \qquad A^{-1} = \frac{1}{v}\begin{pmatrix} 1 & -a & 0 \\ -a & 1+a^2 & -a \\ 0 & -a & 1 \end{pmatrix}.$$

For $b = (c, a^2, a)^T$, a direct multiplication gives

$$b^T A^{-1} b = a^2 + \frac{d^2}{v}.$$

The Schur-complement determinant formula therefore gives

$$\det T_{4,1}(f_\alpha) = v^2\left(1 - a^2 - \frac{d^2}{v}\right) = v^3 - vd^2 = v^3\left(1 - \frac{a^2}{324}\right).$$

Since $v > 0$ and $a > 0$, the strict inequality follows. □

Remark 2.3 (Independent two-block verification). Reflection of coordinates commutes with $T_{4,1}(f_\alpha)$. Its restrictions to the even and odd reflection subspaces are

$$B_+ = \begin{pmatrix} 1+c & a+a^2 \\ a+a^2 & 1+a \end{pmatrix}, \qquad B_- = \begin{pmatrix} 1-c & a-a^2 \\ a-a^2 & 1-a \end{pmatrix}.$$

Their determinants are $(1+a)(v+d)$ and $(1-a)(v-d)$. Their product is $v(v^2-d^2)$, independently recovering (2.4). The defect d enters the two sectors with opposite signs and hence lowers their product by the strictly positive quantity vd^2.

Corollary 2.4 (A rational certificate). *For $\alpha = 1/2$, an admissible function has $a_2 = 1/2$, $a_3 = 1/4$ and $a_4 = 7/48$, and*

$$\det\begin{pmatrix} 1 & 1/2 & 1/4 & 7/48 \\ 1/2 & 1 & 1/2 & 1/4 \\ 1/4 & 1/2 & 1 & 1/2 \\ 7/48 & 1/4 & 1/2 & 1 \end{pmatrix} = \frac{1295}{3072} < \frac{1296}{3072} = \left(1 - \frac{1}{4}\right)^3.$$

The lower bound fails by exactly $1/3072$.

2.3 Moment positivity and propagation to every higher order

The propagation step uses only classical facts, which are stated explicitly to distinguish them from the counterexample calculation. The Marx–Strohhäcker theorem gives

$$\Re\frac{f(z)}{z} > \frac{1}{2} \qquad (z \in \mathbb{D}) \tag{2.5}$$

for every normalized convex function f; see, for example, [2] for the theorem and its stronger generalizations.

Lemma 2.5 (Classical Toeplitz positivity and decreasing innovations). *If f is normalized and convex, every $T_{q,1}(f)$ is positive semidefinite and*

$$0 \le D_q(f) := \det T_{q,1}(f) \le 1.$$

With $D_0 = D_1 = 1$, the positive successive determinant ratios D_{q+1}/D_q are nonincreasing. Once a determinant vanishes, all subsequent determinants vanish.

Proof. By (2.5), the function $P(z) = 2f(z)/z - 1$ has positive real part and $P(0) = 1$. Herglotz representation identifies the coefficients a_{n+1} with the moments of a positive probability measure on the unit circle, making $T_{q,1}(f)$ a Gram matrix. Alternatively, for $0 < r < 1$, the positive weight $\Re P(re^{i\theta})\, d\theta/(2\pi)$ gives the matrix with entries $r^{|j-k|}(T_{q,1})_{jk}$; its limit as $r \uparrow 1$ is positive semidefinite. Its diagonal entries equal one, so Hadamard's determinant inequality gives $D_q \le 1$.

Applying the Hadamard–Fischer inequality to the two overlapping principal submatrices on indices $\{1, \dots, q\}$ and $\{2, \dots, q+1\}$ gives

$$D_{q+1} D_{q-1} \le D_q^2.$$

For positive determinants this is exactly monotonicity of the successive ratios. It is also the usual monotonicity of monic orthogonal-polynomial norms: multiplying a minimizing polynomial by the circle coordinate preserves its norm and produces an admissible competitor at the next degree.

If $T_{q,1}$ is singular, a null vector extended by a zero coordinate has zero quadratic form in the positive semidefinite $T_{q+1,1}$. It is therefore a null vector there too. Induction proves the last assertion. □

Theorem 2.6 (One analytic witness fails at every order $q \ge 4$)**.** *For every $0 < \alpha < 1$ and every integer $q \ge 4$, the same function (2.3) satisfies*

$$0 \le \det T_{q,1}(f_\alpha) \le (1-\alpha^2)^{q-1} \left(1 - \frac{\alpha^2}{324}\right)^{q-3} < (1-\alpha^2)^{q-1}. \tag{2.6}$$

Proof. Put $v = 1 - \alpha^2$ and $r = 1 - \alpha^2/324$. The coefficient calculation gives $D_2 = v$, $D_3 = v^2$ and $D_4 = v^3 r$. Lemma 2.5 gives $D_{k+1}/D_k \le D_4/D_3 = vr$ for $k \ge 3$ while these ratios are positive. Therefore

$$D_q \le D_3 (vr)^{q-3} = v^{q-1} r^{q-3}.$$

If any determinant vanishes, the same bound remains true for all higher orders. Finally $0 < r < 1$ and $q - 3 \ge 1$, giving the strict comparison with the proposed lower bound. □

Corollary 2.7 (The relative failure grows with the order)**.** *For every fixed $0 < \alpha < 1$,*

$$\lim_{q \to \infty} \frac{\det T_{q,1}(f_\alpha)}{(1-\alpha^2)^{q-1}} = 0.$$

Proof. The quotient is nonnegative and, by (2.6), bounded above by $(1 - \alpha^2/324)^{q-3}$, which tends to zero. □

Corollary 2.8 (Classification of the original parameter range)**.** *For $0 < \alpha \le 1$ and $q \ge 2$, the double inequality (2.1) holds for every member of $\mathcal{C}_\alpha$ if and only if $q \in \{2, 3\}$ or $\alpha = 1$. The upper bound $\det T_{q,1}(f) \le 1$ remains true for every allowed (α, q).*

Proof. The cases $q = 2, 3$ are the proved bounds in [1, Theorems 4 and 5]. Every strongly convex function is convex, so Lemma 2.5 gives the upper bound and, when $\alpha = 1$, the lower bound 0. The remaining parameter range is refuted by Theorem 2.6. The upper endpoint is attained by $f(z) = z$; when $\alpha = 1$, the lower endpoint is attained by $f(z) = z/(1-z)$. □

2.4 What this application does and does not establish

The named statement has been kept unchanged: its original function class, parameter range, normalization and constants all enter the counterexample. An exact analytic function supplies the coefficients, so membership is not inferred from a truncated Taylor polynomial. The

computation is rational and symbolic; no finite-precision sign test or unproved asymptotic premise is used. The all-order conclusion closes the integer quantifier as well.

The mechanism is a coefficient defect relative to a geometric moment sequence. The low-order values $a_2 = \alpha$ and $a_3 = \alpha^2$ agree with that sequence, but $a_4 - \alpha^3 = \alpha(1-\alpha^2)/18 \neq 0$. A Schur complement turns this defect into a strictly negative correction to the determinant, and classical prediction-error monotonicity propagates the correction. This is an external disproof, not an implication for the Riemann hypothesis.

Historical priority remains a separate question. The original conjecture is verified in the primary 2021 publication, and targeted searches of subsequent publications and the exact determinant expression did not identify an earlier matching refutation. Such a search does not establish absence from all prior literature. No claim of first discovery, a historically new general principle, or complete satisfaction of a no-prior-duplication requirement is made here. In particular, the universal convex-class estimate $0 \leq \det T_{q,1}(f) \leq 1$ is explicitly presented as a classical corollary, even though related finite-order conjectures appear in [3, 4].

3 The mean error in Mertens' product: a concrete application

Put

$$B_{\mathbb{P}} = \gamma_E - \sum_p \sum_{k\geq 2} \frac{1}{kp^k}, \qquad E(x) = \sum_{p\leq x} \frac{1}{p} - \log\log x - B_{\mathbb{P}},$$

and define

$$\Delta_1(X) = \int_2^X E(x)\,dx, \qquad \Delta_2(X) = \int_2^X E(x)^2\,dx,$$

$$E_3(x) = \frac{1}{\log x} \prod_{p\leq x} (1-p^{-1})^{-1} - e^{\gamma_E}, \qquad J(X) = \int_2^X E_3(x)\,dx.$$

Here and below a nontrivial zero is denoted by $\rho = \beta + i\gamma$, and

$$\Theta = \sup\{\beta : \zeta(\beta + i\gamma) = 0,\ 0 < \beta < 1\}.$$

The precise external question is Question (1) in [5, Section 8]: can the additional regularity assumption in the $\Theta = 1$ case of that paper's Theorem 2 be removed? The original question concerns the sign of $J(X)$, with the normalization above. We retain that function and that normalization.

Theorem 3.1 (Removal of the auxiliary zero-boundary assumption)**.** *If $\Theta = 1$, then*

$$\limsup_{X\to\infty} J(X) = +\infty, \qquad \liminf_{X\to\infty} J(X) = -\infty.$$

No differentiability, regular variation, or sharpness assumption on an auxiliary description of the zero-free region is required. More precisely, under the same hypothesis,

$$\limsup_{X\to\infty} \frac{\Delta_1(X)}{1+\Delta_2(X)} = +\infty, \qquad \liminf_{X\to\infty} \frac{\Delta_1(X)}{1+\Delta_2(X)} = -\infty. \tag{3.1}$$

The proof below is a deduction from the two published estimates stated in Section 3.1. It is not a proof of the Riemann hypothesis. Its application is to the previously stated question of removing an auxiliary hypothesis from a mean-value characterization.

The zero envelope used here is established prior art. In particular, Pintz [9] studies $\inf_\rho\big((1-\beta)\log x + \log|\rho|\big)$; Révész [10] discusses convexification and the resulting concave envelopes. Broucke [11, Introduction and Section 2] reviews these connections. We make no claim to have introduced this envelope or the associated convex-analysis principle. The step used for the present application is a selection of supporting zeros at which the first-moment oscillation exceeds the entire second moment. Priority of this particular application is a separate bibliographic question; the proof does not establish a historical firstness claim.

3.1 The two published analytic inputs

We use the following statements with their quantifiers unchanged.

Lemma 3.2 (Zero-specific oscillation; Zhao, Lemma 10)**.** *For every $\varepsilon > 0$ there are effectively computable constants $C_1(\varepsilon), C_2(\varepsilon)$ with the following property. If $\rho_0 = \beta_0 + i\gamma_0$ is a zero of ζ with*

$$\beta_0 > \tfrac{1}{2} + \varepsilon, \qquad \gamma_0 > C_1(\varepsilon),$$

then for every $X > \gamma_0^{C_2(\varepsilon)}$ there exist $x_+, x_- \in [X, X^{1+\varepsilon}]$ such that

$$\Delta_1(x_+) > \frac{x_+^{\beta_0}}{\gamma_0^{2+\varepsilon} \log x_+}, \qquad \Delta_1(x_-) < -\frac{x_-^{\beta_0}}{\gamma_0^{2+\varepsilon} \log x_-}. \tag{3.2}$$

Lemma 3.3 (Second moment; Zhao, Lemma 11)**.** *Suppose $\eta : [0, \infty) \to (0, 1/2]$ is continuous and nonincreasing, and that $\zeta(s)$ has no zeros in $\operatorname{Re} s > 1 - \eta(|\operatorname{Im} s|)$. Define*

$$\omega_\eta(x) = \inf_{t \geq 1} \{\eta(t) \log x + \log t\}.$$

For every fixed $0 < \varepsilon < 1$,

$$\Delta_2(x) \ll_\varepsilon \frac{x \exp\{-2(1-\varepsilon)\omega_\eta(x)\}}{(\log x)^2}. \tag{3.3}$$

The version published in [5] also allows finitely many exceptional zeros when $\Theta = 1$; that extension is not needed here.

These are imported results, not claims re-proved in this section. Zhao derives the first by adapting Pintz's localized oscillation method [7], and the second from an explicit-formula second-moment bound and Pintz's zero-free-region estimate [8]. In particular, the proof of the first lemma in [5] is a sketch; we have not independently reconstructed every step of that Pintz-method adaptation. Thus the logical dependency of Theorem 3.1 includes the published statements of these two lemmas. Continuity and nonincrease, with no derivative condition, are the hypotheses in the second lemma.

3.2 A supporting-line selection lemma

Lemma 3.4. *Let $\{(a_j, v_j) : j \in \mathcal{I}\}$ be a nonempty family satisfying*

$$a_j > 0, \quad v_j > 0, \quad \inf_j a_j = 0, \quad a_j v_j \geq c > 0,$$

and suppose that only finitely many indices have $v_j \leq V$ for any fixed V. Fix $a_0 > 0$ and set

$$\Omega(u) = \min\left\{a_0 u, \inf_{j \in \mathcal{I}} (a_j u + v_j)\right\}, \qquad u > 0.$$

For each $0 < \kappa < 1$ there are arbitrarily large u and an index j such that

$$\Omega(u) = a_j u + v_j, \qquad a_j u \geq \kappa v_j. \tag{3.4}$$

Along any such sequence with $u \to \infty$,

$$a_j \to 0, \qquad v_j \to \infty, \qquad v_j = o(u).$$

Proof. For fixed u, the auxiliary line bounds the infimum by a_0u; therefore only indices with $v_j \le a_0u$ can minimize it. There are finitely many of them, so the minimum is attained. As the infimum of positive affine functions with positive slopes, Ω is positive, nondecreasing, and concave, hence locally absolutely continuous on $(0,\infty)$. The arithmetic–geometric mean inequality gives

$$a_ju + v_j \ge 2\sqrt{a_jv_ju} \ge 2\sqrt{cu}.$$

The auxiliary line also exceeds a positive constant times $\sqrt{u}$ for all sufficiently large u. Consequently

$$\Omega(u) \gg \sqrt{u} \qquad (u \to \infty). \tag{3.5}$$

On the other hand, $\Omega(u)/u \le a_j + v_j/u$ for every fixed j, and $\inf_j a_j = 0$. Thus

$$\Omega(u) = o(u). \tag{3.6}$$

In particular the auxiliary line is not a minimizer for all sufficiently large u.

Suppose the asserted selection fails. Then for all sufficiently large u, every minimizing actual line satisfies $a_ju < \kappa v_j$. At a point of differentiability of Ω, the slope of every minimizing line equals $\Omega'(u)$. Hence, almost everywhere on a terminal interval,

$$\frac{u\Omega'(u)}{\Omega(u)} = \frac{a_ju}{a_ju + v_j} < \frac{\kappa}{1+\kappa}.$$

Integrating the logarithmic derivative gives

$$\Omega(u) \ll u^{\kappa/(1+\kappa)}.$$

Since $\kappa/(1+\kappa) < 1/2$, this contradicts (3.5).

For a selected pair, $a_j \le \Omega(u)/u \to 0$ and $v_j \le \Omega(u) = o(u)$. If v_j stayed bounded along an infinite subsequence, local finiteness would put those indices in a fixed finite set with strictly positive minimum a_j, a contradiction. This proves the remaining conclusions. □

3.3 One fixed continuous zero-free envelope

Assume now that $\Theta = 1$. Let $\mathcal{Z}$ be the nontrivial zeros with $\gamma > 0$ and $\beta > 3/4$, counted with multiplicity, and put

$$a_\rho = 1 - \beta, \qquad v_\rho = \log\gamma, \qquad a_0 = \tfrac{1}{4}.$$

The familiar zero-free region $\beta \le 1 - c_0/\log(\gamma+2)$ gives $a_\rho v_\rho \ge c > 0$, after decreasing c if necessary. Here the ordinates are bounded away from 1, and each bounded height contains only finitely many zeros. The hypothesis $\Theta = 1$ gives $\inf_{\rho\in\mathcal{Z}} a_\rho = 0$. Thus Lemma 3.4 applies to

$$\Omega(u) = \min\left\{\frac{u}{4}, \inf_{\rho\in\mathcal{Z}}((1-\beta)u + \log\gamma)\right\}. \tag{3.7}$$

For $v \ge 0$ define the running boundary

$$b(v) = \min\left\{\tfrac{1}{4}, \inf_{\substack{\rho\in\mathcal{Z}\\ v_\rho\le v}} a_\rho\right\},$$

where the infimum over an empty set is ignored. This is a positive nonincreasing function, piecewise constant on compact intervals. Directly from the definition,

$$\inf_{v\ge0}\{b(v)u + v\} = \Omega(u). \tag{3.8}$$

Indeed, a record value a_ρ begins at $v = v_\rho$; waiting longer at the same value increases $b(v)u + v$. The initial value $1/4$ gives the auxiliary line at $v = 0$.

Define, once and for all,

$$\eta(t) = \int_0^1 b(\log t + s)\, ds \quad (t \ge 1), \qquad \eta(t) = \eta(1) \quad (0 \le t \le 1). \tag{3.9}$$

This function is positive, continuous, and nonincreasing, and

$$b(\log t + 1) \le \eta(t) \le b(\log t) \qquad (t \ge 1). \tag{3.10}$$

At every zero in $\mathcal{Z}$, $\eta(\gamma) \le b(\log \gamma) \le 1 - \beta$. For the remaining zeros $\beta \le 3/4$, the same inequality follows from $\eta \le 1/4$. Conjugation handles negative ordinates. It follows that the open region $\sigma > 1 - \eta(|t|)$ is zero free.

This construction uses one fixed η, independent of the selected zeros and of u. In particular the implied constant in Lemma 3.3 does not change with the subsequence to be chosen. Equations (3.8)–(3.10) give

$$\Omega(u) - 1 \le \omega_\eta(e^u) \le \Omega(u). \tag{3.11}$$

For the lower bound, write $v = \log t$ and observe that

$$\eta(e^v)u + v \ge b(v+1)u + v = b(v+1)u + (v+1) - 1 \ge \Omega(u) - 1.$$

The upper bound follows by taking infima in $\eta(e^v) \le b(v)$.

3.4 First-moment oscillations dominate the second moment

Fix the numerical choices

$$\kappa = \frac{1}{2}, \qquad \varepsilon = \frac{1}{20}.$$

By Lemma 3.4, there are $u_j \to \infty$ and zeros $\rho_j = \beta_j + i\gamma_j$ for which, with $a_j = 1 - \beta_j$ and $v_j = \log \gamma_j$,

$$\Omega(u_j) = a_j u_j + v_j, \qquad a_j u_j \ge \tfrac{1}{2} v_j, \qquad a_j \to 0, \quad v_j \to \infty, \quad v_j = o(u_j). \tag{3.12}$$

In particular, for all sufficiently large j,

$$\beta_j > \tfrac{1}{2} + \varepsilon, \qquad \gamma_j > C_1(\varepsilon), \qquad e^{u_j} > \gamma_j^{C_2(\varepsilon)}.$$

Lemma 3.2 supplies points $x_{j,+}, x_{j,-} \in [e^{u_j}, e^{(1+\varepsilon)u_j}]$ of the respective signs. Write $w_{j,\pm} = \log x_{j,\pm}$. For either sign, abbreviate $w = w_{j,\pm}$ and

$$A_{j,\pm} = |\Delta_1(e^w)|.$$

Then

$$A_{j,\pm} > \frac{\exp\{w - a_j w - (2+\varepsilon)v_j\}}{w}. \tag{3.13}$$

The fixed-envelope estimate (3.3), the comparison (3.11), and monotonicity of Ω give, with a fixed constant $C > 0$,

$$\Delta_2(e^w) \le C \frac{\exp\{w - 2(1-\varepsilon)(a_j u_j + v_j)\}}{w^2}. \tag{3.14}$$

Divide (3.13) by (3.14). Since $w \le (1+\varepsilon)u_j$,

$$\begin{aligned}\log \frac{A_{j,\pm}}{\Delta_2(e^w)} &\ge 2(1-\varepsilon)a_j u_j - a_j w - 3\varepsilon v_j + \log w - O(1)\\ &\ge (1-3\varepsilon)a_j u_j - 3\varepsilon v_j + \log w - O(1)\\ &\ge \left(\tfrac{1}{2}(1-3\varepsilon) - 3\varepsilon\right) v_j + \log w - O(1)\\ &= \frac{11}{40} v_j + \log w - O(1) \longrightarrow +\infty. \end{aligned} \tag{3.15}$$

If $\Delta_2(e^w) = 0$, the required domination is immediate; in this application it is positive for every $e^w > 2$. Furthermore, since $a_j \to 0$ and $v_j = o(u_j)$, (3.13) implies

$$A_{j,\pm} \ge \frac{\exp\{(1-o(1))u_j\}}{(1+\varepsilon)u_j} \longrightarrow \infty. \tag{3.16}$$

Equations (3.15) and (3.16) prove (3.1).

The distinction from merely changing variables is visible in (3.15): it is an asymptotically divergent ratio between the signed contribution and the complete nonnegative quadratic error. No cancellation conjecture or finite-height zero verification enters this comparison.

3.5 The exact product identity and both signs

The conversion from E to E_3 must retain the prime-power tail

$$R(x) = \sum_{p>x} \sum_{k\ge 2} \frac{1}{kp^k} \ge 0. \tag{3.17}$$

The exact Euler-product identity is

$$e^{-\gamma_E} E_3(x) = \exp\{E(x) - R(x)\} - 1. \tag{3.18}$$

Indeed, expanding the logarithm of the finite product and using the definition of $B_{\mathbb{P}}$ gives

$$\log\left(\frac{e^{-\gamma_E}}{\log x} \prod_{p\le x} (1-p^{-1})^{-1}\right) = E(x) - R(x).$$

The tail is small by an elementary bound:

$$0 \le R(x) \le \frac{1}{2} \sum_{n>x} \frac{1}{n(n-1)} = \frac{1}{2\lfloor x \rfloor} \le \frac{1}{x} \qquad (x \ge 2). \tag{3.19}$$

Here $1/k \le 1/2$ for $k \ge 2$ was applied before summing the geometric series, and replacing primes by all integers only increases the sum. In particular

$$\int_2^X R(x)\,dx \le \log(X/2). \tag{3.20}$$

By Mertens' theorem $E(x) \to 0$; local boundedness then gives $|E(x)| \le M$ for all $x \ge 2$ and some finite M. Taylor's theorem implies

$$e^E - 1 \le E + CE^2 \qquad (|E| \le M)$$

with one fixed C (for example $C = e^M/2$). Using $R \ge 0$ for the upper bound and $e^y - 1 \ge y$ for the lower bound gives the valid two-sided inequalities

$$\Delta_1(X) - \log(X/2) \le e^{-\gamma_E} J(X) \le \Delta_1(X) + C\Delta_2(X). \tag{3.21}$$

At $X = x_{j,-}$, the right-hand side tends to $-\infty$ by (3.15) and (3.16). At $X = x_{j,+}$, the left-hand side tends to $+\infty$: the exponential lower bound in (3.16) exceeds $\log x_{j,+} = O(u_j)$. This proves Theorem 3.1. Since J is continuous, the two unbounded sequences of opposite signs also imply infinitely many sign changes.

Remark 3.5 (Exactness of the nonlinear conversion). The displayed exponential conversion in [5, Section 4] does not include the term $R(x)$. With the definitions of that paper, (3.18) is the exact identity. The negative tail helps the upper bound needed at the negative points; its integral is negligible at the positive points by (3.20). Thus the argument above does not rely on dropping this term or on an incorrect exact equality.

3.6 The full characterization with the tail retained

Corollary 3.6 (The complete mean-value characterization, with the tail retained)**.** *With the original Mertens-product error E_3 and the original integral J,*

$$\mathrm{RH} \quad \Longleftrightarrow \quad J(X) > 0 \quad \textit{for every } X > 2.$$

If RH is false, then

$$\limsup_{X\to\infty} J(X) = +\infty, \qquad \liminf_{X\to\infty} J(X) = -\infty.$$

The source estimates used for the $\Theta < 1$ cases are specified explicitly in the proof; the exponential identity with an omitted prime-power tail is not used.

Proof. We use the exact sandwich (3.21), already established with the prime-power tail retained.

RH implies positivity for every $X > 2$. Rosser and Schoenfeld's Theorem 23, specifically the lower inequality in [6, Eq. (4.10), p. 73], states

$$e^{\gamma_E} \log x < \prod_{p\le x} \frac{p}{p-1} \qquad (0 < x \le 10^8).$$

Thus $E_3(x) > 0$ for $2 \le x \le 10^8$, and hence $J(X) > 0$ for $2 < X \le 10^8$.

For the remaining range we use the quantitative estimate proved at the end of [5, Section 2, following Eq. (19)]. Under RH,

$$\int_X^\infty \frac{\pi(t) - \mathrm{li}(t)}{t^2}\, dt < -\frac{1.73}{\sqrt{X}\log X} \qquad (X \ge 10^8). \tag{3.22}$$

This estimate belongs to that paper's first-moment proof and does not use its later exponential conversion. The exact partial-summation identity [5, Eq. (18)] is

$$\Delta_1(X) = -X\int_X^\infty \frac{\pi(t) - \mathrm{li}(t)}{t^2}\, dt + C_0, \qquad C_0 = 2B_{\mathbb{P}} + 2\log\log 2 - \mathrm{li}(2).$$

In particular,

$$\Delta_1(X) > 1.73\frac{\sqrt{X}}{\log X} - 2 \qquad (X \ge 10^8). \tag{3.23}$$

For completeness, the rough constant bound $C_0 > -2$ needs no numerical prime computation. Indeed,

$$\sum_p \sum_{k\ge 2} \frac{1}{kp^k} \le \sum_{n\ge 2} \left[-\log(1-n^{-1}) - n^{-1}\right] = 1 - \gamma_E,$$

so $B_{\mathbb{P}} \ge 2\gamma_E - 1$. If $t = \log 2$, then

$$\mathrm{li}(2) = \gamma_E + \log t + \sum_{k\ge 1} \frac{t^k}{k\,k!}, \qquad \sum_{k\ge 1} \frac{t^k}{k\,k!} < e^t - 1 = 1.$$

The elementary inequalities $\gamma_E > 1/2$ and $\log t > -1/2$ give

$$C_0 \geq 3\gamma_E - 2 + \log t - \sum_{k\geq 1} \frac{t^k}{k\,k!} > -2.$$

For $X \geq 10^8$, the function $\sqrt{X}/\log X$ is increasing and exceeds 500. Thus (3.23) gives $\Delta_1(X) > \sqrt{X}/\log X$. Also

$$\frac{\sqrt{X}}{\log X} > \log X \qquad (X \geq 10^8).$$

To verify the last inequality, write $u = \log X$: $e^{u/2}/u^2$ is increasing for $u > 4$ and exceeds 1 at $X = 10^8$. Using the lower inequality in (3.21), we obtain

$$e^{-\gamma_E} J(X) > \log X - \log(X/2) = \log 2 > 0 \qquad (X \geq 10^8).$$

This proves the entire RH direction while retaining R.

The case $1/2 < \Theta < 1$. The two inputs are [5, Eq. (20)], which gives for every $\varepsilon > 0$

$$\Delta_1(X) = \Omega_{\pm}(X^{\Theta-\varepsilon}),$$

and [5, Lemma 9], which gives

$$\Delta_2(X) = O_\Theta\left(\frac{X^{2\Theta-1}}{(\log X)^2}\right).$$

Choose $0 < \varepsilon < 1 - \Theta$. Then $\Theta - \varepsilon > 2\Theta - 1 > 0$. Along the positive sequence provided by the first input, the lower inequality in (3.21) tends to $+\infty$, because $\log X = o(X^{\Theta-\varepsilon})$. Along its negative sequence, the upper inequality tends to $-\infty$, because $\Delta_2(X) = o(X^{\Theta-\varepsilon})$. Thus the discarded tail in the original exponential calculation does not change either sign in this case.

The case $\Theta = 1$. Theorem 3.1 gives the two unbounded signs without the additional regularity assumption. By the functional equation, failure of RH is equivalent to $\Theta > 1/2$. The preceding two cases therefore establish the converse and the stated oscillation assertion. □

Remark 3.7 (Scope of the imported estimates). The proof above independently repairs the nonlinear passage to the Mertens product. It imports Rosser–Schoenfeld's finite-range product inequality, Zhao's explicit first-moment bound under RH, and Zhao's first- and second-moment estimates for $1/2 < \Theta < 1$. The $\Theta = 1$ argument retains its separately stated dependency on the published localized oscillation and zero-free-region second-moment lemmas. None of these dependencies is replaced by the incorrect identity obtained by setting R equal to zero.

4 Scope, reproducibility, and historical status

4.1 What the mathematical arguments close

The Toeplitz conclusion is an unconditional refutation of a universally quantified assertion in an identifiable published source. A counterexample is a complete negative answer to that assertion. The analytic witness is defined on the entire disk, not only by a finite coefficient list. The fourth-order determinant can be checked independently of every result in the number-theoretic part of this paper. The higher-order argument also classifies the whole original parameter range, while leaving the problem of a replacement sharp lower bound open here.

The Mertens argument addresses the original integrated product error. Its new deduction from the specified published estimates is the existence of supporting zeros for which the signed first moment dominates the full second moment. The envelope is fixed before the subsequence is chosen, so the implied constants do not depend on a varying auxiliary function. All selected points tend to infinity, and the prime-power tail is bounded before drawing conclusions about the product.

Verification question	Finding
Externally posed target	KLS Conjecture 1 and Zhao Section 8, Question (1), both identified in primary publications.
Original problem retained	The analytic class, matrix normalization, parameters and constants are unchanged in the first target. The second uses the original Mertens product and ordinary integral.
Complete quantifiers	The witness fails for all $q \geq 4$ and all $0 < \alpha < 1$. In the $\Theta = 1$ branch, both signed sequences are unbounded and no additional regularity hypothesis is assumed.
Exact versus numerical	The Toeplitz calculation is a polynomial identity over $\mathbb{Q}$. The analytic application uses estimates and a subsequence argument, not numerical zero data.
Imported theory	Classical convexity, Herglotz positivity and Hadamard–Fischer; for the arithmetic application, Zhao's stated local oscillation and second-moment lemmas and the classical zero-free region.
Novelty and open status	The sources explicitly pose the targets. No matching resolution was identified in the bounded subsequent-literature search. That does not certify continued open status at the time of this note or historical priority of these arguments.

4.2 The relation to the originating prime-circle manuscript

The transferable objects are positive circle moments and signed arithmetic moments. In the first application, circle moments supply a sequence of positive Toeplitz matrices. A coefficient discrepancy that is invisible to the first two determinants is detected by a fourth-order Schur complement, then persists through decreasing prediction errors. In the second application, a convex-analytic description of zero locations is combined with first- and second-moment estimates; the useful conclusion is the divergent ratio in (3.15).

These are specific applications of established mathematical structures. The Cayley map, Schur complements, prediction errors and lower envelopes are not introduced as historically new principles. Their names do not establish a transfer on their own: the transfer is the proved class membership and negative determinant correction in the first case, and the uniform estimates and supporting-line selection in the second.

4.3 Literature checks and remaining limits

The literature check used the original authors, titles, conjecture wording, DOIs, exact determinant constants, and later work on strongly convex Hermitian Toeplitz determinants. An indexed citation list for KLS included eleven later records dated 2022–2026; the inspected nearby papers concerned different function classes or low-order determinants. Searches for Zhao's question included its assumption number, integrated Mertens errors, first/second moments and zero envelopes. No matching refutation or assumption removal was found in the inspected material.

This is a reproducible description of a bounded search, not a proof of absence from the literature. In particular, this paper does not assert that all historical novelty requirements have been certified. The general envelope method is positively identified as prior art in [9, 10, 11]. The original strongly convex witness is credited to [1]. The claims should be judged on the explicit proofs and these stated dependencies, not on an unsupported claim of first discovery.

A An exact arithmetic certificate

The following standalone Python program uses only the standard library and rational arithmetic. It verifies the complete polynomial identity by the permutation formula, separately from the Schur-complement proof. It then checks the rational instance and its exact gap. It is included

in the single `.tex` source so that this note has no external code or figure dependencies. This certificate verifies the arithmetic; Lemma 2.1 supplies the independent analytic membership proof.

```
"""Exact rational certificate for the q=4 KLS 2021 counterexample.

No floating-point arithmetic and no external dependencies are used.
"""
from fractions import Fraction as F
from itertools import permutations

def det(M):
    n = len(M)
    out = F(0)
    for p in permutations(range(n)):
        sign = (-1) ** sum(p[i] > p[j] for i in range(n) for j in range(i + 1, n))
        term = F(sign)
        for i in range(n):
            term *= M[i][p[i]]
        out += term
    return out

def add(a, b):
    c = [F(0)] * max(len(a), len(b))
    for i, x in enumerate(a): c[i] += x
    for i, x in enumerate(b): c[i] += x
    while len(c) > 1 and c[-1] == 0: c.pop()
    return c

def mul(a, b):
    c = [F(0)] * (len(a) + len(b) - 1)
    for i, x in enumerate(a):
        for j, y in enumerate(b): c[i+j] += x*y
    while len(c) > 1 and c[-1] == 0: c.pop()
    return c

def pdet(M):
    n = len(M)
    out = [F(0)]
    for p in permutations(range(n)):
        sign = (-1) ** sum(p[i] > p[j] for i in range(n) for j in range(i+1,n))
        term = [F(sign)]
        for i in range(n): term = mul(term, M[i][p[i]])
        out = add(out, term)
    return out

one, a, a2 = [F(1)], [F(0),F(1)], [F(0),F(0),F(1)]
c = [F(0), F(1,18), F(0), F(17,18)]
M = [[one,a,a2,c],[a,one,a,a2],[a2,a,one,a],[c,a2,a,one]]
actual = pdet(M)
v = [F(1),F(0),F(-1)]
claimed = mul(mul(mul(v,v),v),[F(1),F(0),F(-1,324)])
assert actual == claimed
print("Polynomial identity verified exactly:")
print("det T4 = (1 - alpha^2)^3 (1 - alpha^2/324)")
print("Coefficients [degree: coefficient]:", {i:str(x) for i,x in enumerate(actual) if x})

a = F(1,2)
c = (a + 17*a**3)/18
M = [[F(1),a,a*a,c],[a,F(1),a,a*a],[a*a,a,F(1),a],[c,a*a,a,F(1)]]
D = det(M)
B = (1-a*a)**3
assert c == F(7,48)
assert D == F(1295,3072)
assert B == F(1296,3072)
assert B-D == F(1,3072)
print("alpha=1/2 coefficients:", [str(x) for x in M[0]])
print("determinant:", D)
print("conjectured lower bound:", B)
print("exact lower-bound violation:", B-D)
```

B Reproducibility notes

The document is self-contained as a LaTeX source and may be compiled with `pdflatex` (twice for cross-references) or `latexmk -pdf`. There are no external images, bibliography databases or source-file inputs. All mathematical displays are vector text. The finite certificate has been run successfully with exact rational arithmetic; the main determinant identity was also checked by the reflection-block factorization.

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